

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

30th October 2014

Subject CT4 – Models

Time allowed: Three Hours (10.30 – 13.30 Hrs)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception*
2. *Mark allocations are shown in brackets.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet.*
4. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)**
- i)** Define each of the following examples of a stochastic process
 - a)** a symmetric simple random walk
 - b)** a compound Poisson process (2)
 - ii)** For each of the processes in (i), classify it as a stochastic process according to its state space and the time that it operates on. (2)
 - iii)** For a stochastic process $\{X_t : t \in J\}$, explain the meaning of the following conditions:
 - a)** strict stationarity
 - b)** weak stationarity (3)
- [7]**
- Q. 2)**
- i)** Let X and Y be two processes with rate λ and μ respectively. Prove that the $X + Y$ is also a Poisson process with rate $(\lambda + \mu)$. (2)
 - ii)**
 - a)** Explain what is meant by a Markov jump chain.
 - b)** Describe the circumstances in which the outcome of the Markov jump chain differs from the standard Markov chain with the same transition matrix. (4)
- At an airport there are N adjacent taxi counters, each of which can service taxi booking requests of passengers who exit from the airport. Arrivals of passengers at the counters are assumed to follow a Poisson process with rate λ . The time taken to book a taxi for the passenger at any given counter is assumed to follow an exponential distribution with mean $1/\mu$.
- iii)** Formulate a Markov Jump process for above and specify:
 - a)** the state space; and
 - b)** the transition diagram (3)
 - iv)** State the Kolmogorov forward equations for the process, in component form. (2)
 - v)** Comment on the appropriateness of the assumptions made regarding passenger arrival and the taxi booking process. (2)
 - vi)**
 - a)** Set out the transition matrix of the jump chain associated with the taxi booking process.
 - b)** Determine the probability that all desks are in use before any passenger has completed the taxi booking, given that no passengers have arrived at booking counters at the outset (4)
- [17]**

- Q. 3)** A study is undertaken by railway authorities in respect of the new train engines that they are planning to buy. The train engines being used in a country A are observed for a month and details pertaining to their operating hours, maintenance jobs, In-Yard hours (i.e. not on duty hours) etc are being captured.

The authorities have observed the following:

- A perfectly working Engine moves to a Yard after it has been in operation for a given time interval.
- An Engine currently resting in a Yard is called upon for operation after specific time intervals.
- If an Engine either currently in Operation or put into Operation from Yard develops any fault then is moved to a maintenance facility for repairs.
- After successful repairs, the engine moves to Yard before being pressed into Operation.
- If the fault cannot be resolved the engines are retired from further service.

The authorities are interested in estimating the rate of Operating hours and maintenance downtime for the new engines.

- i) Draw a diagram illustrating a multiple-state model which the investigators could use to make their estimates, using the four states: "In-Yard", "Operation", "Maintenance" and "Retired". (1)
 - ii) Derive from first principles the Kolmogorov differential equation for "Operation" state. (5)
 - iii) Write down the likelihood of the data in terms of the waiting times in each state, the numbers of transitions of each type, and the transition intensities, assuming the transition intensities are constant. (3)
 - iv) Derive the maximum likelihood estimator of the rate "Operation" state. (2)
- [11]**

- Q. 4)** A no claims discount system operates with four levels of discount, 0%, 20%, 40% and 50%. If a policyholder makes no claim during the year he moves up a level of discount (or remains at the maximum level). If he makes one claim during the year he moves down one level of discount (or remains at the minimum level) and if he makes two or more claims he moves down to, or remains at, the minimum level.

The probability for a policyholder making two or more claims in a year is 25%.

The long-term probability of being at the 0% discount level is the same as the long-term probability of being at the 50% discount level.

- i)** Derive the probability of a policyholder making exactly one claim in a given year. (4)
- ii)** Determine the probability that a policyholder at the 20% level this year will be at the 50% level after three years. (2)
- iii)** Estimate the probability that a policyholder at the 0% level this year will be at the 40% level after 20 years, without calculating the associated transition matrix. (3)

Company A has launched a new motor insurance product that provides cover for three years with the total premium for the three years being paid upfront by the policyholder. Policyholder's NCD is applied to the total premium for the three years. Mr. John has purchased this product and the gross annual premium is Rs. 25,000/- for his car. He is currently at 40% level of NCD.

The company still uses the above principle for its NCD system at the renewal of the policy after three years but considers the total no of claims made by the policyholder during the previous three years.

Mr. John made two claims of Rs. 7,500 and Rs. 10,000 during the three years.

- iv)** Determine if Mr. John can save money by not claiming either or both the claims and protecting his NCD assuming that the annual premium charged by the company has reduced by 20% after 3 years. Comment which option gives the best savings to Mr. John. (5)
- [14]**

Q. 5) The Hazard function is given by

$$1/(w-x) \quad \text{where } 0 < x < w;$$

$$\text{and, } V[T(15)] = 533.4$$

where, $T(x)$ is the future life time at exact age x

- i)** Calculate complete expectation of life at age 25. (4)
 - ii)** Comment on the appropriateness of the given form of the hazard function for modelling human mortality over the life time. (1)
- [5]**

- Q. 6)** Table below gives the data for a small sample of patients in a hospital from 2 groups undergoing treatment for lung disease. It shows the time in days until the first death occurs. Observations marked + show the time of leaving for those patients who left hospital alive.

Group 0: 4+, 6, 8+, 9, 11+

Group 1: 3, 5, 5+, 6, 8+

A Cox Proportional Hazards Model

$$\lambda(t;Z) = \lambda_0(t) * \exp(\beta Z)$$

is to be fitted (assuming censoring occurs just after death at the respective duration) to these data where t is the time until death, $\lambda_0(t)$ is the baseline hazard and $z = 0$ for group 0, $z = 1$ for group 1.

- i)** What are the advantages of using semi-parametric Cox Proportional hazard model as against using fully parametric or non-parametric approaches? (2)
 - ii)** Obtain the log partial likelihood for the data considering all possible combinations of d_j deaths out of the $R(t_j)$ at risk at time t_j , given that the events are truly tied at day 6. (5)
 - iii)** An initial estimate of B^* , the maximum likelihood estimate of B , is 1.80. Use an iteration of the Newton-Raphson method to obtain an improved estimate of B^* . (6)
- The Newton-Raphson method uses the fact that if x is a good initial approximation to a root of the equation $f(x) = 0$, then $x^* = x - [f(x)/f'(x)]$ is usually a better approximation.*
- iv)** Using the given estimate of B test the hypothesis that the hazard rate differ by group and state your conclusions. (2)
- [15]**

- Q. 7)** Life insurance companies, investigating the mortality experience of their policyholders, usually count the number of policies in force and the number of policies giving rise to death claims, rather than the number of lives assured in force and the number of lives who die during the period of investigation.

- i)** Briefly discuss why life insurance companies might be following such an approach. Describe the statistical problems that might arise from an investigation based on policies rather than on the lives assured. (3)
 - ii)** If N identical and independent lives aged x exact are observed until age $x + 1$ (or until earlier death), and π_i is the proportion of those lives having exactly i policies ($i = 1, 2, 3, \dots$), then show algebraically how the statistical problems you described in part (i) arise. (4)
- [7]**

Q. 8) $t_1, t_2, \dots, t_n$ are a random sample from an Exponential distribution with parameter u .

We observe N number of parts which are used for manufacturing 1 unit of medical equipment up to time t_s . From the N parts observed we see that N_f parts fail, with failure times $t_1, t_2, \dots, t_f$. The remaining N_w (i.e. N minus N_f parts) items are still working at time t_s .

i) Derive the likelihood estimate of u using the information provided (5)

ii) 50 components are tested for two weeks. 20 of them failing in this time, with an average failure time of 1.2 weeks. What is the mean time till failure? (2)
[7]

Q. 9) A life insurance company proposes to graduate the crude estimates obtained from its investigations before using them for determining premiums.

i) Graduation is said to resolve the conflicting requirements of smoothness and adherence to data. Explain:

a) how this conflict arises, and

b) how the process of graduation resolves the conflict (4)

ii) Describe how smoothness is ensured when mortality rates are graduated using each of the following methods:

a) fitting a parametric formula

b) graphical graduation (2)

iii) The company has decided to fit the mortality rates for the ages(x) 30 to 34 by using the straight line $a+b \cdot x$ to the crude values of $\log_e(q_x/p_x)$ with estimated parameters of $a = -10.5$ and $b = 0.1162$. The actual number of deaths which were observed for the respective ages during last year is provided in the table below.

Age (x)	30	31	32	33	34
No of deaths	245	276	313	323	339

The initial exposed to risk at all ages may be assumed to be 2,50,000.

Test that the model used to fit the crude values of $\log_e(q_x/p_x)$ could be considered a good fit given actual death information. (5)
[11]

Q. 10) The government of a country wishes to model recent and prospective mortality changes. Discuss the requirements of a suitable model and the challenges faced in applying it to mortality data? (6)
[6]
